AGGREGATE SHOCKS AND DEBT MANAGEMENT I

Fiscal and Monetary Policy 2026

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PLAN

- How should the government change taxes and debt in response to changes in purchases?
- For example: should it finance war spending in the same way as public education?
- This is a normative question, but to the extent real world governments behave according to prescriptions of models we will see in class, it will also be a positive theory of government debt and taxes.

PLAN

- We will study optimal taxation and debt policy in two environments:
  - incomplete markets: the government can buy or sell only a limited set of securities, often only a single risk-free security.
  - complete markets: the government can buy or sell claims contingent on all possible states of the world.
- First study a reduced form model (Barro, 1979), then proceed to optimal taxation in a competitive equilibrium (Lucas and Stokey, 1983).

REDUCED FORM MODELS

INCOMPLETE MARKETS

- We study the following problem based on Barro (1979):
  - the government uses taxes τ_t to finance *stochastic* purchases g_t;
  - $D(\tau_t)$ is the deadweight loss of taxes;
  - the government wants to finance g_t in a way that minimizes the expected value of present discounted deadweight loss of taxes:

$$\min_{\{\tau_t\}_{t=0}^{\infty}} \mathbb{E}_0 \sum_{t=0}^{\infty} \left[\left(\frac{1}{1+r} \right)^t D(\tau_t) \right]$$

  where $1/(1+r)$ is the government's discount factor.

INCOMPLETE MARKETS

- Assume that g_t is governed by an N state Markov chain.
- The state of the world is given by s_t that follows a Markov chain with a transition probability matrix P:

$$P_{ij} = P\left\{s_{t+1} = \bar{s}_j \mid s_t = \bar{s}_i\right\}.$$

- Government spending $\{g_t\}$ obeys

$$g_t = \begin{cases} g_1 & \text{if } s_t = \bar{s}_1 \\ \vdots & \\ g_N & \text{if } s_t = \bar{s}_N. \end{cases}$$

INCOMPLETE MARKETS

- The government can issue only one period, non-state contingent debt (incomplete markets) at the price $1/(1+r)$, equal to the discount rate.
- Government problem:

$$\min_{\{\tau_t\}_{t=0}^{\infty}} \mathbb{E}_0 \sum_{t=0}^{\infty} \left[\left(\frac{1}{1+r} \right)^t D(\tau_t) \right]$$

$$\text{s.t.} \quad g_t + b_t = \frac{1}{1+r} b_{t+1} + \tau_t$$

and, in addition, we assume

$$\mathbb{E}_0 \left[\left(\frac{1}{1+r} \right)^t b_t^2 \right] \leq \infty.$$

RECURSIVE FORMULATION OF THE GOVERNMENT'S PROBLEM

- Write the problem recursively:

$$V(b,g) = \min_{\tau} D(\tau) + \frac{1}{1+r}\mathbb{E}\left[V\left(b',g'\right) \mid g\right]$$
$$\text{s.t.} \quad g + b = \frac{1}{1+r}b' + \tau$$

where x denotes current variables and x' next-period variables.

THE FOC EQUATES MARGINAL DISTORTION TO THE SHADOW VALUE OF DEBT

- Substitute the constraint and differentiate with respect to τ:

$$D'(\tau) = \mathbb{E}\left[\frac{\partial V(b', g')}{\partial b'} \,\middle|\, g\right]$$

- The envelope condition gives us $V'(b, g) = D'(\tau)$, so

$$V'(b', g') = D'(\tau').$$

THE EULER EQUATION: MARGINAL DISTORTION TODAY = EXPECTED MARGINAL DISTORTION TOMORROW

- Combining the FOC with the envelope condition:

$$D'(\tau_t) = \mathbb{E}_t\left[D'(\tau_{t+1})\right]$$

- Tax smoothing: spread out distortionary effects of taxes across time.

INCOMPLETE MARKETS

- There are two special cases we will consider in more detail:
  - Special case 1 - no uncertainty (perfect foresight):

$$\tau_t = \tau_{t+1}$$

  - Special case 2 - quadratic deadweight loss $(D(\tau_t) = a\tau_t^2 + \gamma)$:

$$\tau_t = \mathbb{E}_t[\tau_{t+1}]$$

SPECIAL CASE 1: PERFECT FORESIGHT

- With no uncertainty about future g_t (perfect foresight), taxes are constant.
- τ (a single value) is chosen to satisfy the budget constraint:

$$\tau = \frac{r}{1+r}\left[\sum_{t=0}^{\infty}\left(\frac{1}{1+r}\right)^{t} g_t + b_0\right]$$

- Debt absorbs all fluctuations in government spending:

$$b_{t+1} - b_t = r\left(b_t - b_0\right) + \left(1+r\right) g_t - r\sum_{s=0}^{\infty}\left(\frac{1}{1+r}\right)^{s} g_s.$$

SPECIAL CASE 1: PERFECT FORESIGHT

- If $g_t = g$ is constant, then
$$\tau = g + \frac{r}{1+r} b_0, \quad b_t = b_0;$$
debt remains constant at its initial level.
- If $g_0 = g + \epsilon$ and $g_t = g$ for all $t \geq 1$ then
$$\tau = g + \frac{r}{1+r} (\epsilon + b_0), \quad b_t = b_0 + \epsilon \text{ for } t \geq 1;$$
debt increases by ϵ and stays elevated forever.
- There is no debt target: b_0 shifts the constant tax level, but absent new spending surprises debt simply stays where it is.
- What matters for changes in debt is transitory fluctuations in g_t, not the average level of g_t.
- Intuition valid also with trend growth of GDP and government spending (see Barro's paper).

SPECIAL CASE 2 - QUADRATIC DEADWEIGHT LOSS

- What does the condition

$$\tau_t = \mathbb{E}_t\left[\tau_{t+1}\right]$$

  tell us about the optimal tax policy?

- τ_t is a random walk:

$$\begin{aligned}\tau_{t+1} - \tau_t &= \tau_{t+1} - \mathbb{E}_t\left[\tau_{t+1}\right] \\ &= \tau_{t+1} - \left(\tau_{t+1} - \epsilon_{t+1}\right) \\ &= \epsilon_{t+1}\end{aligned}$$

  where ϵ_{t+1} is a martingale-difference innovation.

SPECIAL CASE 2 - QUADRATIC DEADWEIGHT LOSS

- Since $\mathbb{E}_{t-1}[\tau_t] = \tau_{t-1}$, the innovation $\tau_t - \mathbb{E}_{t-1}[\tau_t]$ moves only because of news about expected future g_t.
- In a persistent but non-absorbing high-spending state, observing the state survive one more period is itself new information.
- To see this, use the budget constraint together with $\mathbb{E}_t[\tau_{t+1}] = \tau_t$ to get

$$\tau_t = \frac{r}{1+r} b_t + \frac{r}{1+r} \sum_{s=0}^{\infty} \left(\frac{1}{1+r}\right)^s \mathbb{E}_t[g_{t+s}].$$

so

$$\tau_t - \mathbb{E}_{t-1}[\tau_t] = \frac{r}{1+r} \sum_{s=0}^{\infty} \left(\frac{1}{1+r}\right)^s \left(\mathbb{E}_t[g_{t+s}] - \mathbb{E}_{t-1}[g_{t+s}]\right)$$

SPECIAL CASE 2 - QUADRATIC DEADWEIGHT LOSS

- The size of the jump in τ_t depends on the size of the change in expected future g_t:
  - If there is a permanent shift in g_t, $\mathbb{E}_t\left[g_{t+s}\right] - \mathbb{E}_{t-1}\left[g_{t+s}\right] = \epsilon$ for all s, taxes change by exactly the same amount ϵ.
  - If the change is temporary, $\mathbb{E}_t\left[g_{t+s}\right] - \mathbb{E}_{t-1}\left[g_{t+s}\right] = \epsilon$ only for $s = 0$, taxes change by $\epsilon r/\left(1+r\right)$.
- Remaining needed resources raised through debt issuance:
  - Debt remains constant for a permanent change in g_t.
  - Debt changes by ϵ for a temporary change in g_t.
- Implication: finance education by taxes, finance war by debt.

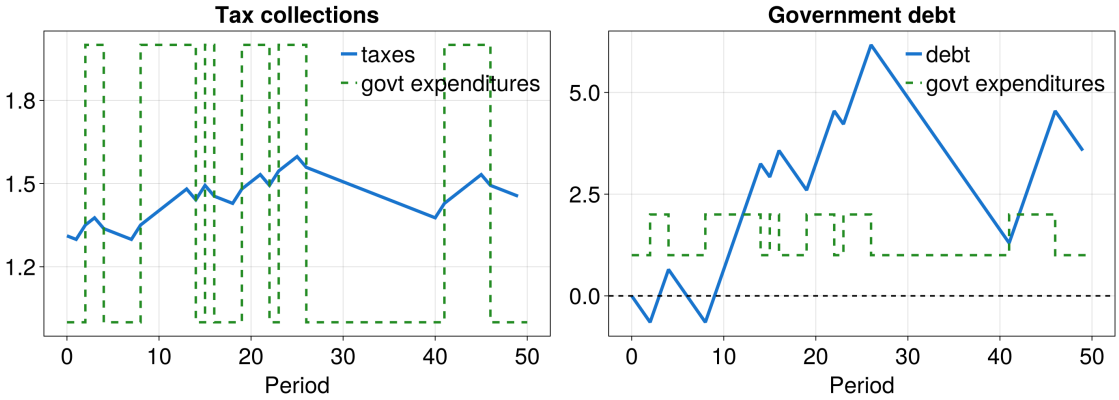
SPECIAL CASE 2 - QUADRATIC DEADWEIGHT LOSS

- Debt is the residual financing object:

$$b_{t+1} = (1 + r)(g_t + b_t - \tau_t).$$

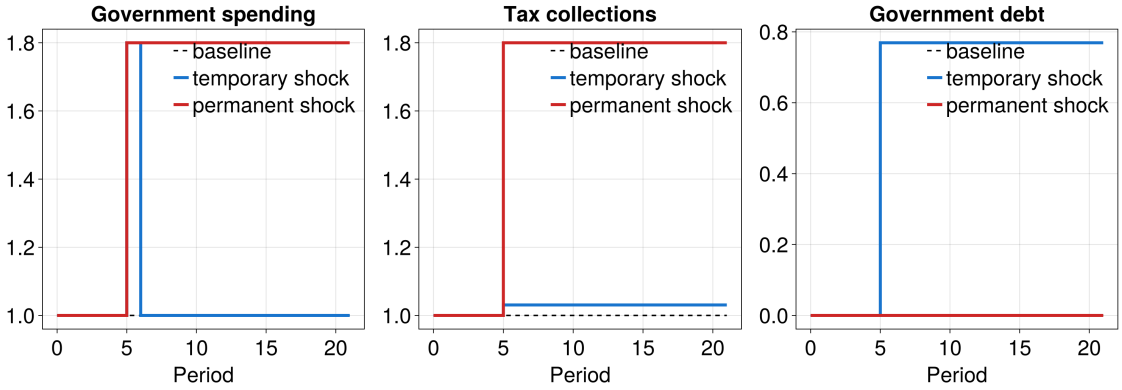
- There is nothing that acts as a force that would push debt to a particular level:
  - No debt target;
  - No "raise taxes if debt is too high" rule;
  - Debt need not revert to any stable level; it can drift for long periods as tax smoothing shifts shocks into future debt service.

NUMERICAL EXAMPLE: BARRO INCOMPLETE MARKETS



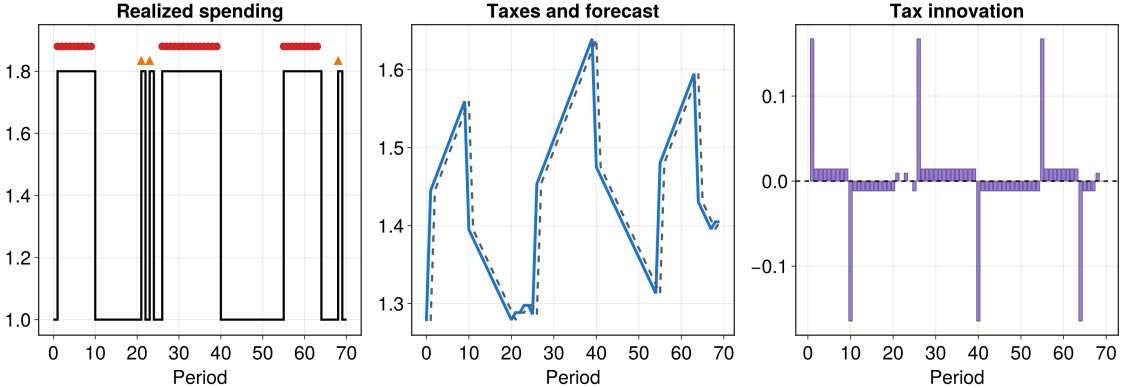
Quadratic tax loss: $D(\tau) = a\tau^2 + \gamma$. Parameters: $\beta = 0.96$, $g_t \in \{1, 2\}$, $P_{12} = 0.2$, $P_{21} = 0.4$, initial debt $b_0 = 0$.

TEMPORARY VS PERMANENT SPENDING SHOCKS



Temporary shock: $\Delta\tau = \epsilon r/(1+r)$ and $\Delta b = \epsilon$. Permanent shock: $\Delta\tau = \epsilon$ and debt is unchanged.

THREE-STATE MARKOV SIMULATION: TAX INNOVATIONS



Dashed gray line: last period's forecast $\mathbb{E}_{t-1}[\tau_t] = \tau_{t-1}$. Purple bars: tax innovation $\tau_t - \mathbb{E}_{t-1}[\tau_t]$. Red circles mark persistent-shock periods; orange triangles mark temporary-shock periods. Both shock states have the same current size, $g_t = \bar{g} + \epsilon$, but state 2 is non-absorbing, so surviving one more period in that state is itself news.

LUCAS AND STOKEY (1983)

SETUP

- We study the following problem based on Lucas and Stokey (1983):
  - the representative household has preferences over consumption and labor (leisure);
  - production technology is linear in labor, there is no capital;
  - the government uses distortionary tax rate τ_t on labor income to finance *stochastic* purchases g_t (we assume g_t is a Markov process);
  - the households and the government can trade a full set of Arrow securities (state-contingent claims).
  - the government wants to finance g_t in a way that maximizes the welfare of the representative household.
- General equilibrium.

SETUP

- Price of securities no longer exogenous.
- The government cares about welfare of the representative household.
- Think of the problem in two steps:
  1. for given government policies there exists some competitive equilibrium;
  2. the government picks policies that result in the "best" equilibrium.
- Key difference: policies not only have to satisfy the government's budget constraint, but also household's optimality conditions.
- Primal approach: we will look directly for allocations that maximize welfare and then think of policies that implement that.

THE PRIMAL APPROACH IN FOUR STEPS

1. Write down the household problem, the firm problem, and feasibility conditions;
2. Derive the household first order conditions;
3. Substitute out prices and policies $\Rightarrow$ implementability constraint: a single restriction on allocations that summarizes private optimality;
4. Formulate the Ramsey problem: maximize welfare subject to implementability + resource constraints.

NOTATION: CURRENT STATE VS. HISTORY

- s_t denotes the current exogenous state at date t.
- $s^t \equiv (s_0, \ldots, s_t)$ denotes the entire history up to date t.
- An object indexed by s^t may in principle depend on the whole history:

$$c\left(s^t\right),\ \ell\left(s^t\right),\ \tau\left(s^t\right),\ a\left(s^{t+1}\right),\ q\left(s^{t+1} \mid s^t\right).$$

- The notation $s^{t+1} \geq s^t$ means that s^{t+1} is a successor history that extends the current history by one state realization.
- Since g_t is Markov, probabilities depend on the history only through the last state:

$$\pi\left(s^{t+1} \mid s^t\right) = \pi\left(s_{t+1} \mid s_t\right).$$

- Later we show that under the Ramsey allocation many objects become history independent, so they can be written as functions of s_t alone.

STEP 1: THE HOUSEHOLD PROBLEM WITH ARROW SECURITIES

- Household chooses consumption, labor supply and portfolio of Arrow securities to maximize expected utility:

$$
\begin{aligned}
& \max \mathbb{E}_0 \sum_{t=0}^{\infty} \beta^t u\left(c\left(s^t\right), \ell\left(s^t\right)\right) \\
\text{s.t.} \quad \forall s^t \quad & c\left(s^t\right) + \sum_{s^{t+1} \geq s^t} q\left(s^{t+1} \mid s^t\right) a\left(s^{t+1}\right) \\
& = \left(1 - \tau\left(s^t\right)\right) w\left(s^t\right) \ell\left(s^t\right) + a\left(s^t\right) \\
& a\left(s^0\right) = a_0
\end{aligned}
$$

where $u(\cdot)$ is the utility function, $c(\cdot)$ consumption and $\ell(\cdot)$ labor.

STEP 1: COMPLETE MARKETS COLLAPSE TO A SINGLE BUDGET CONSTRAINT

- Impose the natural debt limit: in each state s^t, the household cannot borrow more than the present value of its minimum future after-tax income:

$$a\left(s^t\right) \geq -\bar{A}\left(s^t\right).$$

- This is the tightest borrowing limit that does not shrink the budget set relative to Arrow-Debreu.
- Iterate the sequential budget constraint forward T periods. The natural debt limit ensures that the terminal term $\sum_{s^{T+1}} q^0\left(s^{T+1}\right) a\left(s^{T+1}\right) \geq 0$, where $q^0\left(s^t\right)$ is the time-0 Arrow-Debreu price.
- Passing to the limit, the present-value budget constraint follows:

$$\sum_{t=0}^{\infty} \sum_{s^t} q^0\left(s^t\right) c\left(s^t\right) \leq \sum_{t=0}^{\infty} \sum_{s^t} q^0\left(s^t\right)\left(1-\tau\left(s^t\right)\right) w\left(s^t\right) \ell\left(s^t\right) + q^0\left(s^0\right) a_0.$$

STEP 1: THE HOUSEHOLD PROBLEM WITH A SINGLE CONSTRAINT

- With locally nonsatiated preferences, the present-value budget constraint binds at the optimum, so the household problem becomes:

$$\max \sum_{t=0}^{\infty} \sum_{s^t} \beta^t \pi\left(s^t\right) u\left(c\left(s^t\right), \ell\left(s^t\right)\right)$$

$$\text{s.t.} \sum_{t=0}^{\infty} \sum_{s^t} q^0\left(s^t\right) c\left(s^t\right) = \sum_{t=0}^{\infty} \sum_{s^t} q^0\left(s^t\right)\left(1-\tau\left(s^t\right)\right) w\left(s^t\right) \ell\left(s^t\right) + q^0\left(s^0\right) a_0$$

- This is nice because there is only one constraint.
- Under the natural debt limit, this present-value problem is equivalent to the original sequential Arrow-securities problem.

STEP 2: HOUSEHOLD FIRST ORDER CONDITIONS

- Write down the Lagrangian and differentiate with respect to $c\left(s^t\right)$ and $\ell\left(s^t\right)$ to get the first order conditions:

$$\left[c\left(s^t\right):\right] \ \beta^t \pi\left(s^t\right) u_c\left(c\left(s^t\right), \ell\left(s^t\right)\right) = \lambda q^0\left(s^t\right)$$
$$\left[\ell\left(s^t\right):\right] \ \beta^t \pi\left(s^t\right) u_\ell\left(c\left(s^t\right), \ell\left(s^t\right)\right) = -\lambda q^0\left(s^t\right)\left(1-\tau\left(s^t\right)\right) w\left(s^t\right)$$

- Note: the Lagrange multiplier λ is the same for all states s^t.
- To simplify notation

$$\left[c\left(s^t\right):\right] \ \beta^t \pi\left(s^t\right) u_c\left(s^t\right) = \lambda q^0\left(s^t\right)$$
$$\left[\ell\left(s^t\right):\right] \ \beta^t \pi\left(s^t\right) u_\ell\left(s^t\right) = -\lambda q^0\left(s^t\right)\left(1-\tau\left(s^t\right)\right) w\left(s^t\right)$$

- Under standard concavity and interiority assumptions, these FOCs characterize the household optimum.

FIRM PROBLEM

- Output in this economy is produced by a representative price-taking firm.
- Linear production function that uses labor as the only input:

$$y\left(s^t\right) = A\ell\left(s^t\right).$$

- The firm chooses labor to maximize profits:

$$\max_{\ell(s^t)} A\ell\left(s^t\right) - w\left(s^t\right)\ell\left(s^t\right).$$

- In a competitive equilibrium,

$$w\left(s^t\right) = A.$$

GOVERNMENT FEASIBILITY

- A competitive equilibrium must also satisfy the government's budget constraint in each state.
- For the primal approach, it is enough to work with the aggregate resource constraint:

$$A\ell\left(s^t\right) = g\left(s^t\right) + c\left(s^t\right).$$

- With market clearing and the firm's zero-profit condition, Walras' law makes the government budget constraint redundant.

Competitive equilibrium

A competitive equilibrium given government policies $\tau\left(s^t\right)$ and $g\left(s^t\right)$ is a set of allocations $\left(c\left(s^t\right), \ell\left(s^t\right), a\left(s^{t+1}\right)\right)$ and prices $\left(q\left(s^{t+1} \mid s^t\right), w\left(s^t\right)\right)$ such that:

1. Given prices, allocations solve the household problem;
2. Given prices, allocations solve the firm problem;
3. The government budget constraint is satisfied in each state;
4. All markets clear in each state.

- This bundles together the objects we have just introduced and is enough for our purposes.

STEP 3: SUBSTITUTE OUT PRICES AND POLICIES (NECESSITY)

- Necessity: if $\left(c\left(s^t\right), \ell\left(s^t\right)\right)$ is part of a competitive equilibrium, then it satisfies the implementability constraint.
- Since $q^0\left(s^0\right) = 1$, the time-0 consumption FOC implies $\lambda = u_c\left(s^0\right)$.
- From the FOCs, replace $q^0\left(s^t\right)$ in the budget constraint:

$$\text{LHS:} \quad \sum_{t,s^t} q^0\left(s^t\right) c\left(s^t\right) = \frac{1}{u_c\left(s^0\right)} \sum_{t,s^t} \beta^t \pi\left(s^t\right) u_c\left(s^t\right) c\left(s^t\right)$$

$$\text{RHS:} \quad \sum_{t,s^t} q^0\left(s^t\right)\left(1-\tau\left(s^t\right)\right) w\left(s^t\right) \ell\left(s^t\right) = -\frac{1}{u_c\left(s^0\right)} \sum_{t,s^t} \beta^t \pi\left(s^t\right) u_\ell\left(s^t\right) \ell\left(s^t\right)$$

- Multiplying both sides by $u_c\left(s^0\right)$ gives the implementability constraint:

$$\sum_{t=0}^{\infty} \sum_{s^t} \beta^t \pi\left(s^t\right)\left[u_c\left(s^t\right) c\left(s^t\right) + u_\ell\left(s^t\right) \ell\left(s^t\right)\right] = u_c\left(s^0\right) a_0 \qquad \square$$

STEP 3: THE IMPLEMENTABILITY CONSTRAINT

- The implementability constraint

$$\sum_{t=0}^{\infty} \sum_{s^t} \beta^t \pi\left(s^t\right)\left[u_c\left(s^t\right) c\left(s^t\right) + u_\ell\left(s^t\right) \ell\left(s^t\right)\right] = u_c\left(s^0\right) a_0$$

  summarizes household optimality conditions and the budget constraint.
- Prices, taxes, and debt have been eliminated. The constraint involves only allocations and the initial condition a_0.
- Under the natural debt limit, it summarizes the sequential household problem as well.
- The government can only choose allocations that satisfy this constraint.

THE IMPLEMENTABILITY CONSTRAINT IS SUFFICIENT

- Sufficiency: if an interior allocation $\left(c\left(s^t\right), \ell\left(s^t\right)\right)$ satisfies the implementability constraint and the resource constraint, then there exist prices, taxes, and debt such that it is a competitive equilibrium.
- **Proof.** Given an allocation, construct the equilibrium:
  1. Set wages: $w\left(s^t\right) = A$ (from the firm problem).
  2. Set taxes: $\left(1 - \tau\left(s^t\right)\right) = -\frac{u_\ell\left(s^t\right)}{u_c\left(s^t\right)} \frac{1}{A}$ (from the intratemporal FOC).
  3. Set Arrow prices: $q\left(s^{t+1} \mid s^t\right) = \beta\, \pi\left(s_{t+1} \mid s_t\right) \frac{u_c\left(s^{t+1}\right)}{u_c\left(s^t\right)}$.
- These choices make the firm FOC and the household intratemporal/intertemporal FOCs hold.

THE IMPLEMENTABILITY CONSTRAINT IS SUFFICIENT (CONT.)

- We still need Arrow-security holdings that satisfy the sequential household budget constraint at each history s^t.
- Define

$$x\left(s^t\right) \equiv \sum_{j=0}^{\infty} \sum_{s^{t+j} \geq s^t} \beta^j \pi\left(s^{t+j} \mid s^t\right)\left[u_c\left(s^{t+j}\right) c\left(s^{t+j}\right) + u_\ell\left(s^{t+j}\right) \ell\left(s^{t+j}\right)\right]$$

- By construction, $x\left(s^t\right)$ satisfies the recursion

$$x\left(s^t\right) = u_c\left(s^t\right) c\left(s^t\right) + u_\ell\left(s^t\right) \ell\left(s^t\right) + \beta \sum_{s^{t+1} \geq s^t} \pi\left(s^{t+1} \mid s^t\right) x\left(s^{t+1}\right).$$

THE IMPLEMENTABILITY CONSTRAINT IS SUFFICIENT (CONT.)

- Set $a\left(s^t\right) = x\left(s^t\right)/u_c\left(s^t\right)$. Using $u_\ell\left(s^t\right) = -u_c\left(s^t\right)\left(1-\tau\left(s^t\right)\right)A$ and the Arrow-price formula,

$$c\left(s^t\right) + \sum_{s^{t+1} \geq s^t} q\left(s^{t+1} \mid s^t\right) a\left(s^{t+1}\right) = \left(1-\tau\left(s^t\right)\right) A\ell\left(s^t\right) + a\left(s^t\right),$$

  so the household sequential budget constraint holds.
- The implementability constraint gives $x\left(s^0\right) = u_c\left(s^0\right) a_0$, so the initial condition is satisfied.

THE IMPLEMENTABILITY CONSTRAINT IS SUFFICIENT (CONT.)

- Set government Arrow liabilities equal to household claims: $b\left(s^t\right) = a\left(s^t\right)$, so the Arrow-security market clears.
- Using the resource constraint $A\ell\left(s^t\right) = c\left(s^t\right) + g\left(s^t\right)$ and the household budget constraint:

$$g\left(s^t\right) + b\left(s^t\right) = \tau\left(s^t\right) A\ell\left(s^t\right) + \sum_{s^{t+1} \geq s^t} q\left(s^{t+1} \mid s^t\right) b\left(s^{t+1}\right),$$

  which is exactly the government budget constraint.
- Households optimize, firms optimize, markets clear, and the government budget constraint holds.
- Therefore the allocation is part of a competitive equilibrium. □
- The implementability constraint + resource constraint fully characterize the set of competitive equilibrium allocations.

STEP 4: THE RAMSEY PROBLEM

- The government chooses allocations $c\left(s^t\right), \ell\left(s^t\right)$ to maximize the household's utility

$$\sum_{t=0}^{\infty} \sum_{s^t} \beta^t \pi\left(s^t\right) u\left(c\left(s^t\right), \ell\left(s^t\right)\right)$$

subject to the implementability constraint:

$$\sum_{t=0}^{\infty} \sum_{s^t} \beta^t \pi\left(s^t\right) \left[u_c\left(s^t\right) c\left(s^t\right) + u_\ell\left(s^t\right) \ell\left(s^t\right)\right] = u_c\left(s^0\right) a_0$$

and resource constraints:

$$\forall s^t \quad A\ell\left(s^t\right) = g\left(s^t\right) + c\left(s^t\right)$$

STEP 4: FULL COMMITMENT

- Full commitment: the government announces its state-contingent plans at time 0 and cannot change them later.
- Everyone else trusts the government and knows it will stick to its plans.

THE PSEUDO-UTILITY FUNCTION W

- Define the pseudo-utility function:

$$W\left(c\left(s^t\right), \ell\left(s^t\right), \Phi\right) \equiv u\left(c\left(s^t\right), \ell\left(s^t\right)\right) + \Phi\left[u_c\left(s^t\right) c\left(s^t\right) + u_\ell\left(s^t\right) \ell\left(s^t\right)\right]$$

  where Φ is the multiplier on the implementability constraint.

- This absorbs the implementability constraint into the objective. The Ramsey problem becomes:

$$\max \sum_{t=0}^{\infty} \sum_{s^t} \beta^t \pi\left(s^t\right) W\left(c\left(s^t\right), \ell\left(s^t\right), \Phi\right) - \Phi\, u_c\left(s^0\right) a_0 \quad \text{s.t.} \quad \forall s^t: \; A\ell\left(s^t\right) = g\left(s^t\right) + c\left(s^t\right)$$

- The only constraints left are the resource constraints — one for each state s^t.

FIRST ORDER CONDITIONS VIA W

- For $s^t > s^0$, the FOCs with respect to $c\left(s^t\right)$ and $\ell\left(s^t\right)$ are:

$$W_c\left(s^t\right) = \mu\left(s^t\right)$$
$$W_\ell\left(s^t\right) = -\mu\left(s^t\right)A$$

  where $\mu\left(s^t\right)$ is the multiplier on the resource constraint for state s^t.

- Written out:

$$u_c\left(s^t\right) + \Phi\left[u_{cc}\left(s^t\right)c\left(s^t\right) + u_c\left(s^t\right) + u_{c\ell}\left(s^t\right)\ell\left(s^t\right)\right] = \mu\left(s^t\right)$$
$$u_\ell\left(s^t\right) + \Phi\left[u_{c\ell}\left(s^t\right)c\left(s^t\right) + u_\ell\left(s^t\right) + u_{\ell\ell}\left(s^t\right)\ell\left(s^t\right)\right] = -\mu\left(s^t\right)A$$

Φ IS THE ONLY LINK BETWEEN STATES

- For each state s^t, we have three equations in three unknowns $c\left(s^t\right), \ell\left(s^t\right), \mu\left(s^t\right)$:
  - two FOCs: $W_c\left(s^t\right) = \mu\left(s^t\right), W_\ell\left(s^t\right) = -\mu\left(s^t\right) A$;
  - one resource constraint: $A\ell\left(s^t\right) = g\left(s^t\right) + c\left(s^t\right)$.
- Φ is the only link between different states s^t.
- Once we know Φ, allocations for each s^t are determined separately.
- For two different histories with the same current state s_t, the solution must be the same!

IMPLICATIONS

- What are the implications for taxes?
- From household first order condition we have

$$\left(1-\tau\left(s^t\right)\right) w\left(s^t\right)=-\frac{u_\ell\left(s^t\right)}{u_c\left(s^t\right)}$$

and linear production technology implies $w\left(s^t\right)=A$ so

$$1-\tau\left(s^t\right)=-\frac{u_\ell\left(c\left(s^t\right), \ell\left(s^t\right)\right)}{u_c\left(c\left(s^t\right), \ell\left(s^t\right)\right)} \frac{1}{A}.$$

- We have solved for $c\left(s^t\right), \ell\left(s^t\right)$, we can get $\tau\left(s^t\right)$!

IMPLICATIONS

- Since for two different histories with the same current state s_t, consumption and labor are the same – taxes must be the same!
- Conclusion: very different from Barro (1979) – no history dependence.
- If government purchases are i.i.d., taxes are i.i.d. as well.
- Contrast with the incomplete markets case: taxes were a random walk.

TAXES ARE NOT FULLY SMOOTHED IN GENERAL

- Unlike Barro (1979), taxes are not generally constant — they depend on the current state.
- When g_t is high, the government may want to increase labor supply to finance it without a large drop in consumption.

EXAMPLE: SEPARABLE PREFERENCES WITH LINEAR DISUTILITY OF LABOR

- Let $u(c, \ell) = \frac{c^{1-\gamma}}{1-\gamma} - \psi\ell$.
- The pseudo-utility function is

$$W\left(c\left(s^t\right), \ell\left(s^t\right), \Phi\right) = \frac{c\left(s^t\right)^{1-\gamma}}{1-\gamma} - \psi\ell\left(s^t\right) + \Phi\left[c\left(s^t\right)^{1-\gamma} - \psi\ell\left(s^t\right)\right]$$

- The FOCs $W_c = \mu\left(s^t\right)$, $W_\ell = -\mu\left(s^t\right)A$ become:

$$\left[1 + \Phi(1-\gamma)\right] c\left(s^t\right)^{-\gamma} = \mu\left(s^t\right)$$
$$\left[1 + \Phi\right]\psi = \mu\left(s^t\right)A$$

EXAMPLE

- For each state $s^t > s^0$ we have:

$$\frac{1+\Phi}{[1+\Phi(1-\gamma)]A}\psi = c\left(s^t\right)^{-\gamma}$$

  so consumption is constant.

- Taxes satisfy

$$1-\tau\left(s^t\right) = \frac{\psi}{c\left(s^t\right)^{-\gamma}}\frac{1}{A}:$$

  since $c\left(s^t\right)$ is constant, taxes are constant as well.

- With these preferences households supply labor fully elastically, fluctuations in g_t are absorbed by changes in production.

LESSONS

- Remember: the above result is not general.
- What is general is that allocations and taxes are history independent — they are fully determined by the current state s_t.
- What about the optimal path of debt?

ARROW PRICES ARE PINNED DOWN BY ALLOCATIONS

- Using the consumption FOCs together with $q^0\left(s^{t+1}\right) = q^0\left(s^t\right)\, q\left(s^{t+1} \mid s^t\right)$:

$$q\left(s^{t+1} \mid s^t\right) = \beta\, \pi\left(s_{t+1} \mid s_t\right) \frac{u_c\left(s^{t+1}\right)}{u_c\left(s^t\right)}$$

- Since allocations depend only on s_t, prices depend only on (s_t, s_{t+1}).
- Once we solve the Ramsey problem for allocations, all prices are determined.

TIMING WITHIN DATE t

- Current history s^t is observed.
- The Arrow claim chosen at $t-1$ for the realized history s^t pays $b\left(s^t\right)$ at date t.
- At the same date t, taxes $\tau\left(s^t\right) A\ell\left(s^t\right)$ are collected and government spending $g\left(s^t\right)$ is paid.
- The government then sells the new menu $\left\{b\left(s^{t+1}\right)\right\}_{s_{t+1}}$ for date $t+1$ and receives its date-t market value $\sum_{s_{t+1}} q\left(s^{t+1} \mid s^t\right) b\left(s^{t+1}\right)$.

THE GOVERNMENT BUDGET CONSTRAINT PINS DOWN PORTFOLIO VALUE

- The government budget constraint for each history s^t:

$$g\left(s^t\right) + b\left(s^t\right) = \tau\left(s^t\right) A\,\ell\left(s^t\right) + \sum_{s_{t+1}} q\left(s^{t+1} \mid s^t\right) b\left(s^{t+1}\right)$$

- Here $b\left(s^t\right)$ is the payoff due today from last period's portfolio choice, while $\left\{b\left(s^{t+1}\right)\right\}_{s_{t+1}}$ is the menu sold today for tomorrow.
- The term $\sum_{s_{t+1}} q\left(s^{t+1} \mid s^t\right) b\left(s^{t+1}\right)$ is the date-t market value of that newly issued continuation portfolio.
- So this equation is same-date accounting under the timing convention above, not a separate law of motion for debt.
- Once $g\left(s^t\right)$, $\tau\left(s^t\right)$, $\ell\left(s^t\right)$, and $q\left(s^{t+1} \mid s^t\right)$ are known from the Ramsey solution, it pins down the market value today; the full payoff menu is recovered next from $x\left(s^t\right)$.

DEFINE MARGINAL-UTILITY-ADJUSTED DEBT $x\left(s^t\right) \equiv u_c\left(s^t\right) b\left(s^t\right)$

- Multiply the government budget constraint by $u_c\left(s^t\right)$ and substitute Arrow prices:

$$x\left(s^t\right) = u_c\left(s^t\right)\left[\tau\left(s^t\right) A \ell\left(s^t\right) - g\left(s^t\right)\right] + \beta \sum_{s_{t+1}} \pi\left(s_{t+1} \mid s_t\right) x\left(s^{t+1}\right)$$

- Under the Ramsey solution, allocations and Arrow prices depend only on the current state s_t, so the right-hand side depends on s^t only through s_t.
- Hence the inherited marginal-utility-adjusted payoff is also Markov: $x\left(s^t\right) = x\left(s_t\right)$. In vector form:

$$\vec{x} = \vec{h} + \beta \Pi \vec{x}$$

where $h(s) \equiv u_c(s)\left[\tau(s) A \ell(s) - g(s)\right]$ and Π is the Markov transition matrix.

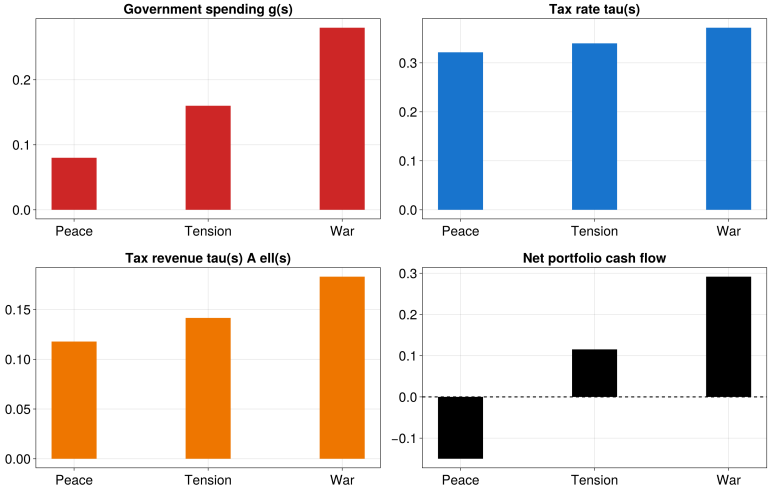
RECOVERING THE NEXT-PERIOD PAYOFF MENU

- Solve $\vec{x} = (I - \beta \Pi)^{-1} \vec{h}$ and recover the payoff promised for any realized state s from

$$b(s) = \frac{x(s)}{u_c(s)}.$$

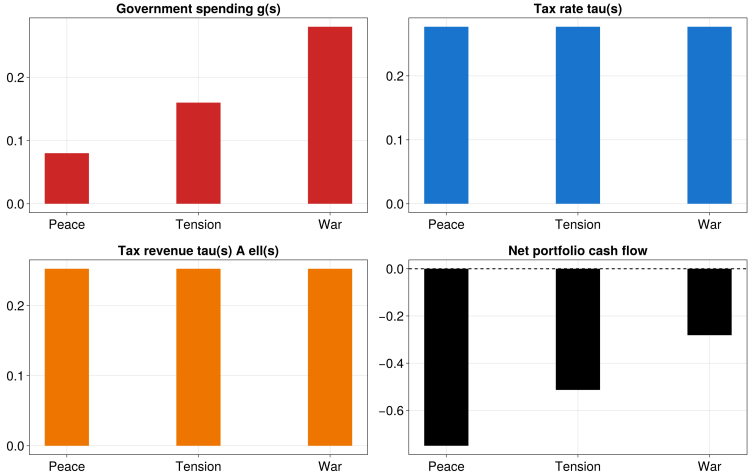
- Interpretation in Markov notation: $b(s_t)$ is the inherited payoff due today, while $\{b(s_{t+1})\}_{s_{t+1}}$ is the menu chosen today for tomorrow's possible states.
- Its date-t market value is $V(s_t) \equiv \sum_{s_{t+1}} q(s_{t+1} \mid s_t)\, b(s_{t+1})$, and $x(s_t) = u_c(s_t)\, b(s_t)$ is the marginal-utility-adjusted payoff.
- Each promised payoff depends only on the successor state s_{t+1}, not on the full history s^t: the issued menu is history independent.

A FINITE-STATE LUCAS-STOKEY EXAMPLE



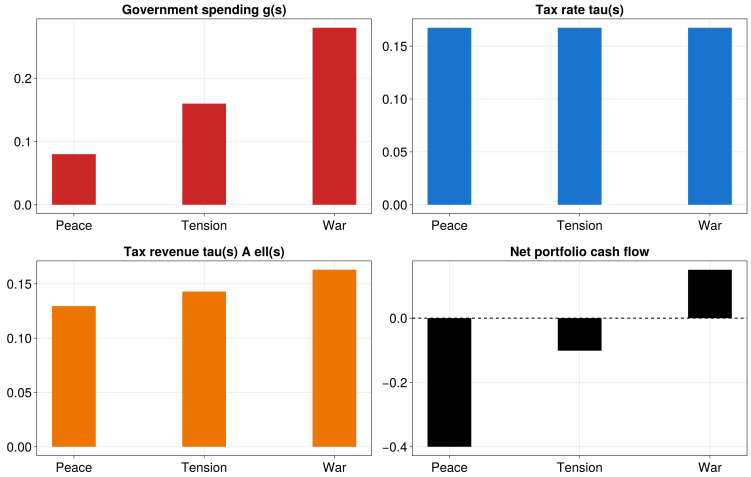
Three-state Markov example with $u(c, \ell) = \log c + \psi \log(1 - \ell)$ and target peace-state payoff $b(\text{peace}) = 0.15$. The four panels show government spending, the tax rate, tax revenue, and net portfolio cash flow $-b(s)$, where positive values mean receipts by the government.

ALTERNATIVE EXAMPLE: QUASI-LINEAR UTILITY



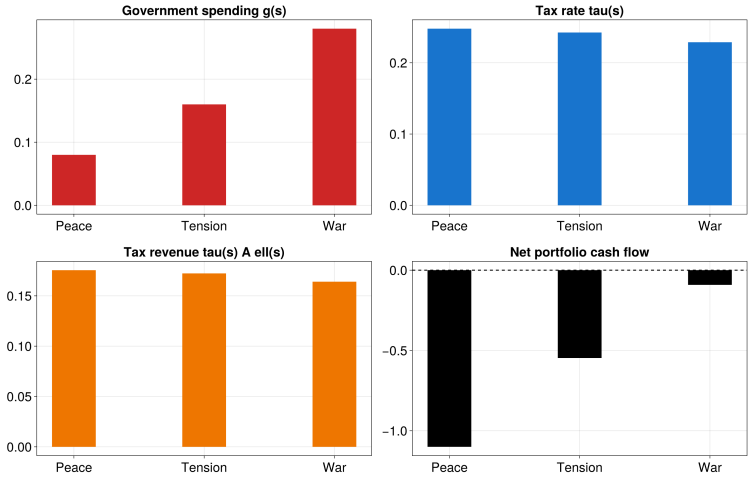
With $u(c, \ell) = c - \psi \ell^{1+\chi}/(1+\chi)$ and high labor curvature $\chi = 8$, u_c is constant. Tax rates are flat across states, while net portfolio cash flow adjusts to absorb the fiscal shock.

ALTERNATIVE EXAMPLE: LOG UTILITY, LINEAR LABOR



With $u(c, \ell) = \log c - \psi\ell$, the tax rate is constant across states, while tax revenue and net portfolio cash flow respond to g.

ALTERNATIVE EXAMPLE: GHH UTILITY

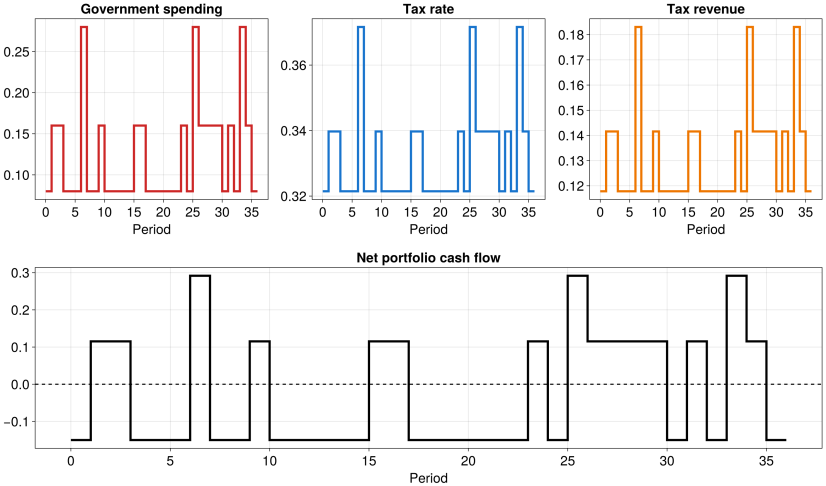


With $u(c, \ell) = \log\big(c - \kappa \ell^{1+\nu}/(1+\nu)\big)$, the tax formula becomes $1 - \tau(s) = \kappa \ell(s)^{\nu}/A$. In this calibration, high g lowers tax rates and raises net portfolio receipts.

STATE-CONTINGENT DEBT AS INSURANCE

- The government chooses a portfolio of state-contingent positions — a separate payoff for each possible next-period state s_{t+1}.
- In the examples, the net portfolio cash flow is more positive in high-g states and more negative in low-g states.
- Effect: the government hedges fiscal shocks, helping keep taxes smoother across states.
- Contrast with incomplete markets: without state-contingent debt, the government cannot hedge $\Rightarrow$ taxes drift as a random walk.

HISTORY INDEPENDENCE IN A SIMULATED PATH



A simulated path from the same three-state Markov chain. Whenever the economy revisits a state, the same spending realization, tax rate, tax revenue, and net portfolio cash flow recur.

TIME INCONSISTENCY AND PORTFOLIO COMMITMENT

THE TIME-0 FOC IS DIFFERENT

- The FOC for $c\left(s^0\right)$ at $t = 0$ has an extra term compared to $t \geq 1$:

$$W_c\left(s^0\right) \underbrace{-\Phi\, u_{cc}\left(s^0\right) a_0}_{\text{extra term}} = \mu\left(s^0\right)$$

$$W_\ell\left(s^0\right) \underbrace{-\Phi\, u_{c\ell}\left(s^0\right) a_0}_{\text{extra term}} = -\mu\left(s^0\right) A$$

- This term comes from differentiating the inherited term $u_c\left(s^0\right) a_0$ in the implementability constraint.
- So period 0 is special: the government in office inherits today's debt burden, while later governments inherit a debt position that period 0 can restructure.

THE CAPITAL LEVY ON INITIAL DEBT

- The extra term gives the time-0 government an incentive to manipulate $u_c\left(s^0\right)$ to reduce the effective burden of inherited debt a_0.
  - For example: policies that raise $c\left(s^0\right)$ lower $u_c\left(s^0\right)$, making debt cheaper to repay.
  - This is a capital levy: expropriating initial debt holders through current taxes and allocations that change $u_c\left(s^0\right)$.
- In Ramsey terms, this relaxes future tax distortions.
- But this is only the date-0 issue. Lucas-Stokey's main question is different: how should debt be restructured so later governments want to continue the plan?

HOW LUCAS-STOKEY DEFINE TIME CONSISTENCY

- In the paper, the date-0 government chooses the current tax, announces future taxes, and restructures debt to be inherited at date 1.
- At date 1, the successor government must honor that inherited debt, but it is otherwise free to choose the current tax, announce later taxes, and restructure debt again.
- The Ramsey plan is time-consistent if, after every realized history, the successor government voluntarily re-chooses the same continuation.
- So future tax announcements alone are not the commitment device. The relevant inherited object is the debt position left to the successor.

WHAT THE DATE-0 PLANNER IS TRYING TO DO

- In the time-consistency exercise, date 0 cannot directly bind future tax rates.
- What it can choose is the debt position inherited by date 1.
- Goal: leave each successor government with inherited debt for which the intended Ramsey continuation solves that government's own problem.
- In our recursive notation, the inherited object is

$$x\left(s^{t+1}\right) = u_c\left(s^{t+1}\right) b\left(s^{t+1}\right).$$

CONTINUATION PLANNER AT HISTORY s^t

- After a realized history s^t, the inherited state is $x\left(s^t\right) \equiv u_c\left(s^t\right) b\left(s^t\right)$.
- Given this state, the date-t government chooses current labor and the menu $\left\{x\left(s^{t+1}\right)\right\}_{s^{t+1} \geq s^t}$ left to successors.
- The recursive implementability constraint is

$$x\left(s^t\right) = u_c\left(s^t\right)\left(A\ell\left(s^t\right) - g\left(s^t\right)\right) + u_\ell\left(s^t\right)\ell\left(s^t\right) + \beta \sum_{s_{t+1}} \pi\left(s_{t+1} \mid s_t\right) x\left(s^{t+1}\right).$$

- Given inherited $x\left(s^t\right)$, the date-t government solves its own Ramsey problem from that state.

TIME-0 PLANNER: INHERITS b_0, NOT $x\left(s^0\right)$

- At date 0, the government inherits goods debt b_0.
- It does not inherit a predetermined $x\left(s^0\right)$, because

$$x\left(s^0\right) = u_c\left(s^0\right) b_0$$

  depends on the current allocation.
- Its recursive implementability constraint is

$$\begin{aligned} u_c\left(s^0\right) b_0 = u_c\left(s^0\right)\left(A\ell\left(s^0\right) - g\left(s^0\right)\right) + u_\ell\left(s^0\right) \ell\left(s^0\right) \\ + \beta \sum_{s_1} \pi\left(s_1 \mid s_0\right) x\left(s^1\right). \end{aligned}$$

- So period 0 is special: it can manipulate the inherited burden $u_c\left(s^0\right) b_0$ and choose the menu $\left\{x\left(s^1\right)\right\}_{s^1 \geq s^0}$ left to date 1.

CHOOSING PROMISED CONTINUATION STATES HISTORY BY HISTORY

- Fix a current history s^t. The continuation planner chooses one promise $x\left(s^{t+1}\right)$ for each successor history $s^{t+1} \geq s^t$.
- The first order condition with respect to each promise is

$$V_x\left(x\left(s^{t+1}\right), s^{t+1}\right) = \Psi\left(s^t\right) \qquad \forall s^{t+1} \geq s^t,$$

  where $\Psi\left(s^t\right)$ is the multiplier on the recursive implementability constraint at history s^t.
- So all successor histories branching from the same current history share the same shadow value $\Psi\left(s^t\right)$.
- To make the plan time-consistent, the planner picks those promises so each date-$(t+1)$ government inherits the state for which the intended continuation solves its own problem.
- This is the recursive version of choosing the restructured debt left to the successor in Lucas-Stokey.

WHAT THE PLANNER COMMITS TO

- In our complete-markets sequential notation, time 0 chooses the menu $\left\{b\left(s^1\right)\right\}_{s^1 \geq s^0}$, or equivalently the inherited states $\left\{x\left(s^1\right) = u_c\left(s^1\right) b\left(s^1\right)\right\}_{s^1 \geq s^0}$.
- This is the recursive counterpart of Lucas-Stokey's "rich enough" maturity/state-contingent debt structure.
- This is the commitment: debt is binding, but future tax choices are not.
- After one particular history s^1 is realized, the continuation planner at that history takes that inherited value $x\left(s^1\right)$ as given.
- It can choose future promises $\left\{x\left(s^2\right)\right\}_{s^2 \geq s^1}$, but it cannot change the inherited object $x\left(s^1\right)$.
- So the commitment technology is the inherited debt portfolio, not a direct command over successors' tax rates.

HOW DEBT MAKES THE PLAN TIME-CONSISTENT

- Date 0 chooses the inherited state left to each successor history.
- It chooses those states so that, after any realized s^1, the intended continuation solves the date-1 government's own continuation problem.
- Successor governments therefore re-optimize into the Ramsey continuation, rather than obeying an externally imposed tax plan.
- This is Lucas-Stokey's result: debt commitments can substitute for direct commitment to future tax rates when the debt menu is rich enough.
- In recursive notation, the portfolio does this by pinning down the inherited state variable at each successor history.

SUMMARY: THE LUCAS-STOKEY RESULT

- An exogenous time-0 debt position creates a capital-levy motive at date 0.
- Lucas-Stokey's main result is different: with a rich enough set of debt instruments, successor governments continue the optimal tax policy because it is optimal for them to do so.
- The key variable is $x(s) = u_c(s)\ b(s)$: marginal-utility-adjusted debt.
- The date-0 planner is not really trying to bind future taxes; it is choosing inherited debt states so that reproducing the Ramsey continuation is each successor's own optimum.
- Without complete markets (Aiyagari, Marcet, Sargent, Seppälä, 2002), the multiplier Φ becomes time-varying and history-dependent — the government cannot fully commit via the debt portfolio.